

Breast Cancer Detection: Wisconsin Diagnostic Dataset

ISA 640 – Capstone Project

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Motivation:

- **Breast cancer** is one of the most common and **life-threatening** diseases globally.
- **Early detection** significantly improves survival rates and treatment outcomes.
- **Machine learning** provides a fast, reliable, and data-driven **diagnosis** approach.



Breast Cancer Numbers in the U.S:

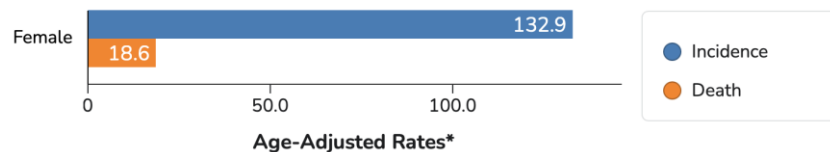
Incidents Rate
per 100,000
Women is 132.9

Incident Count is
279,731

Population
168,308005

Incidence and death rates

Females had much higher rates of getting [breast cancer](#) than dying from breast cancer.



*Age-adjusted rates are per 100,000 standard population.



Introduction:

Problem:

Predict whether a **breast tumor** is **benign** or **malignant**.

Objectives:

- Apply multiple ML models.
- Evaluate performance and interpret features.
- Integrate interpretability to support clinical decision-making.

Methodology:

Data Description:

Wisconsin Diagnostic Breast Cancer Dataset (WDBC)

- 569 samples and 30 numerical diagnostic features.
- Target: (M= malignant, B= benign)
- Sources: UCI Machine Learning Repository

Features Description:

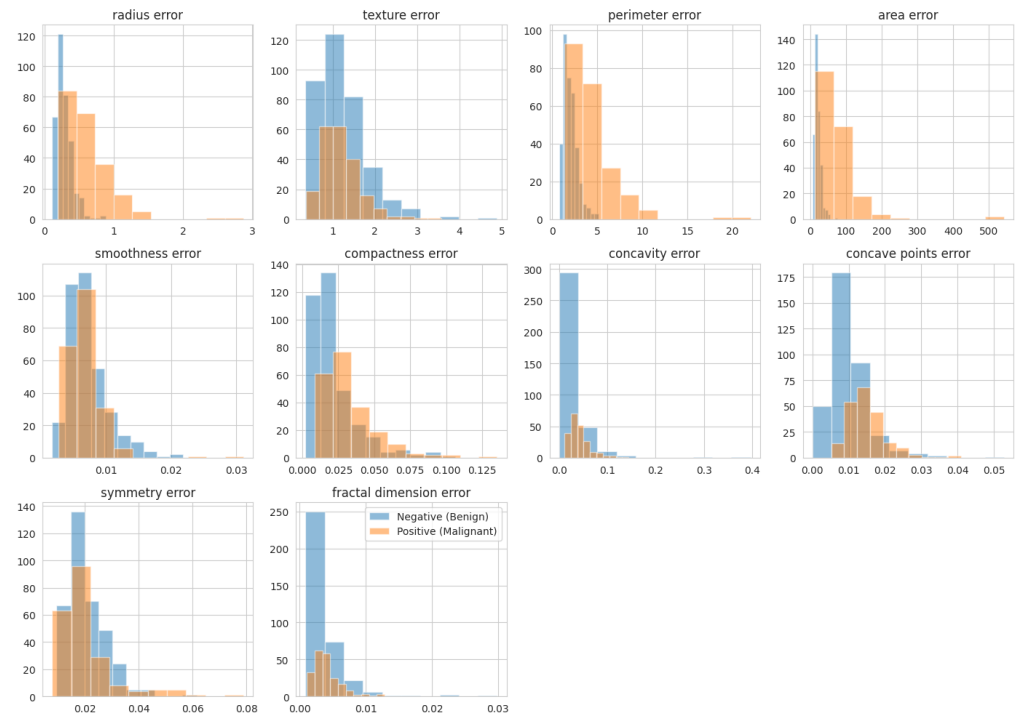
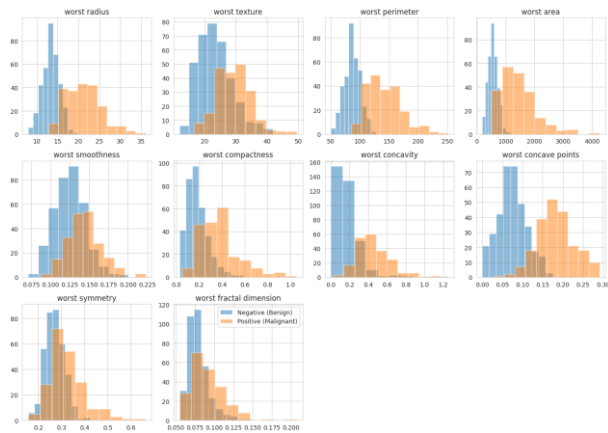
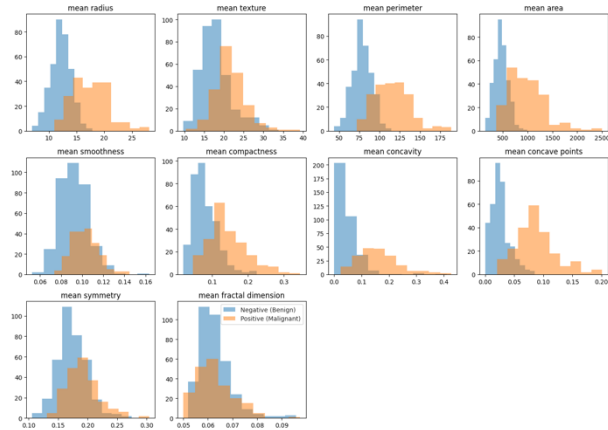
For Each Cell:

- Radius (mean of distances from center to points on the perimeter)
- Texture (standard deviation of gray-scale values)
- Perimeter
- Area
- Smoothness (local variation in radius lengths)
- Compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
- Concavity (severity of concave portions of the contour)
- Concave points (number of concave portions of the contour)
- Symmetry
- Fractal dimension ("coastline approximation" - 1)

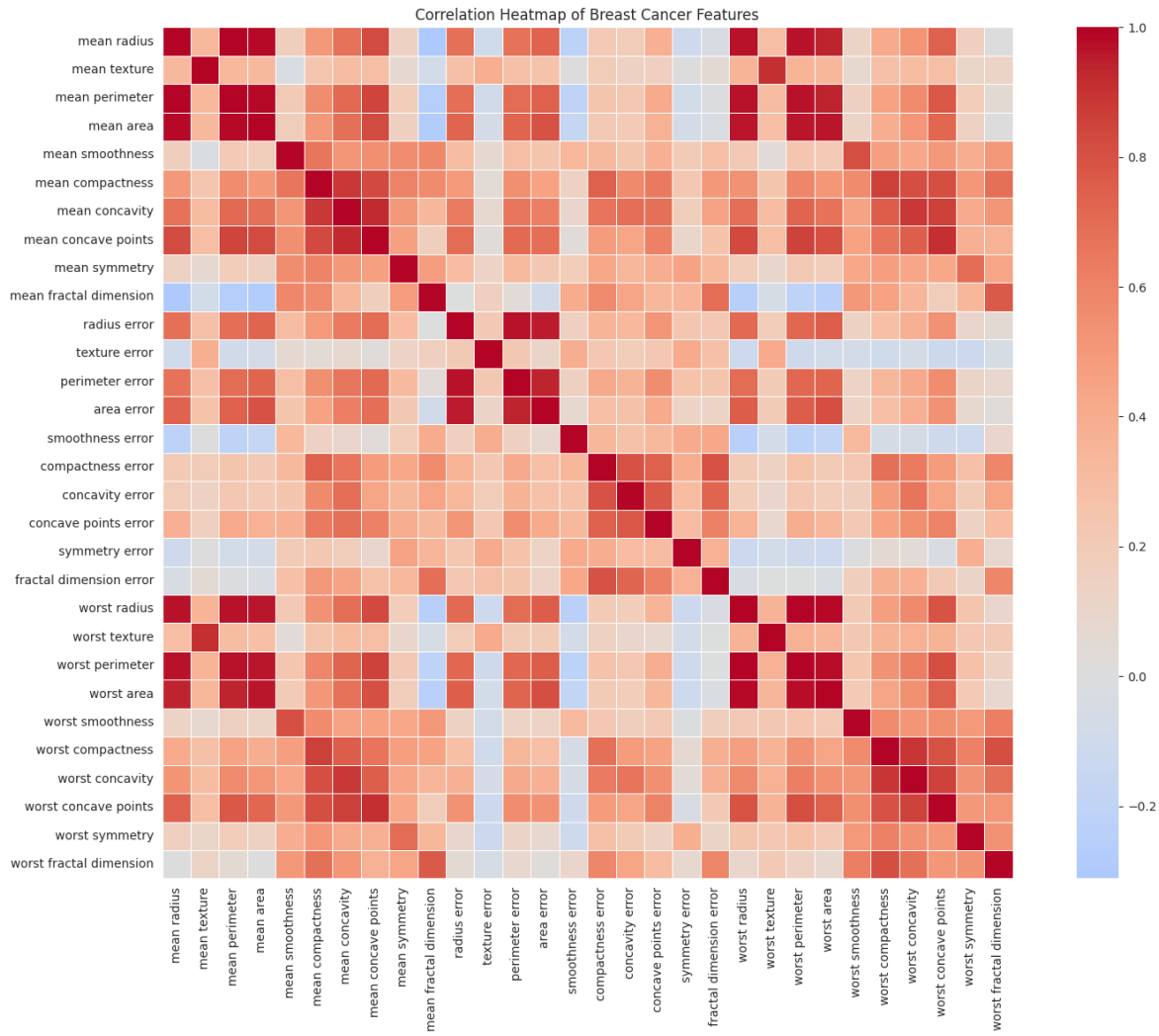
EDA:

- Histograms comparing malignant vs benign tumors
- Correlation heatmap
- Exploratory feature importance
- PCA for dimensionality reduction

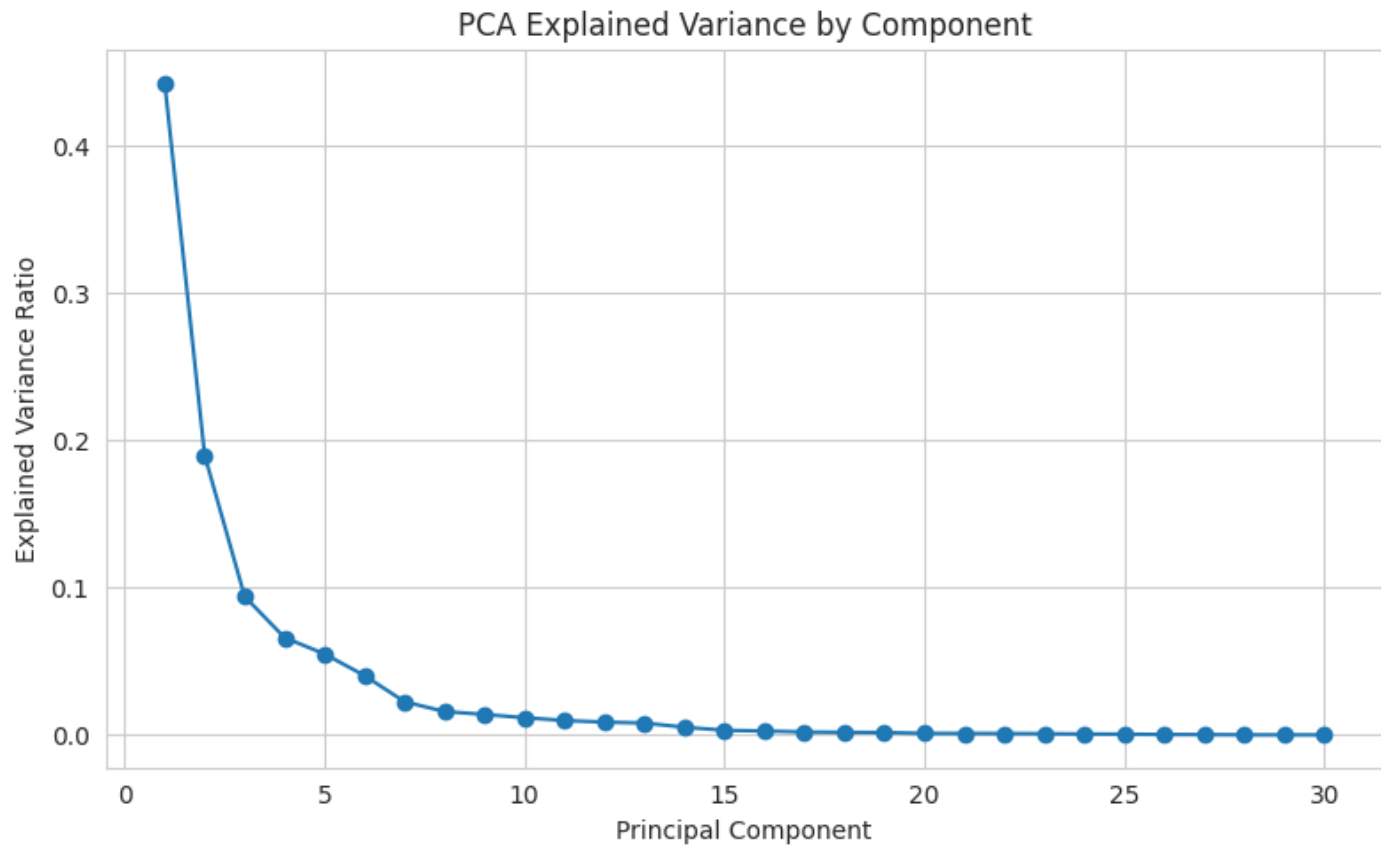
Class Distribution:



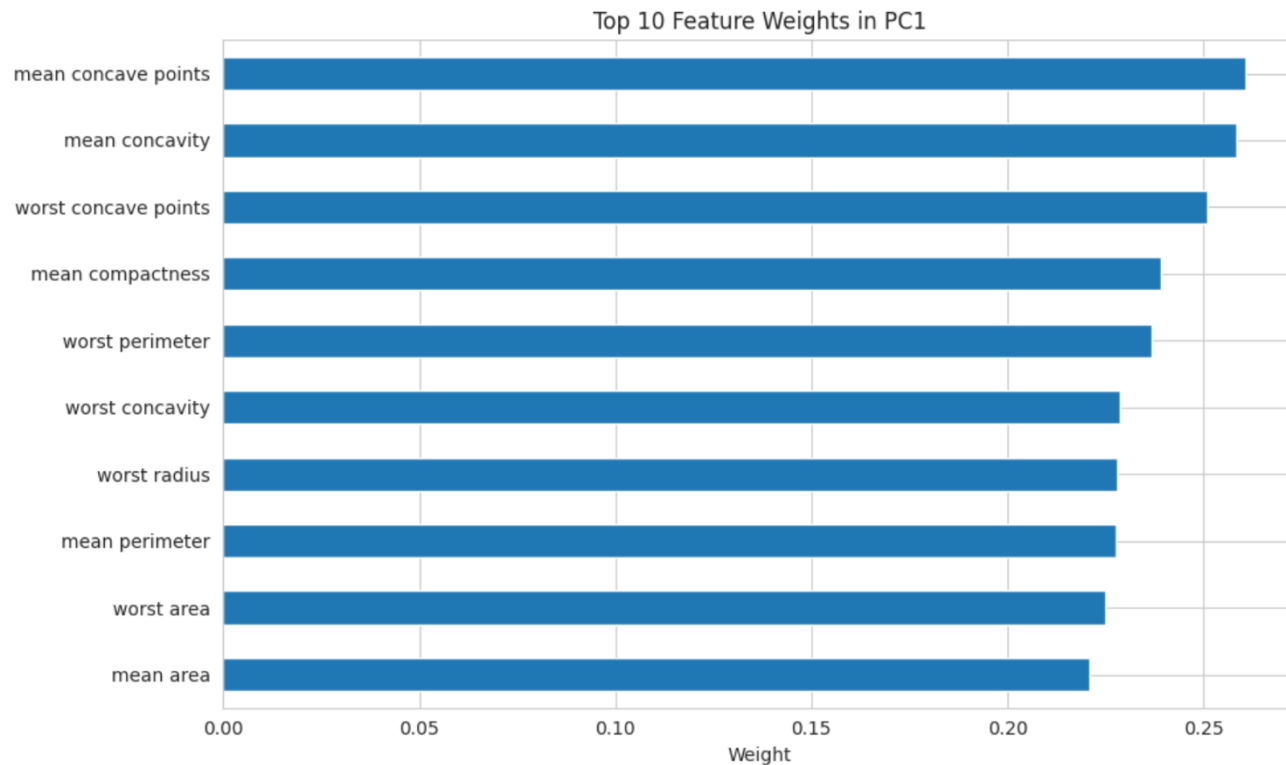
Correlation Heatmap:



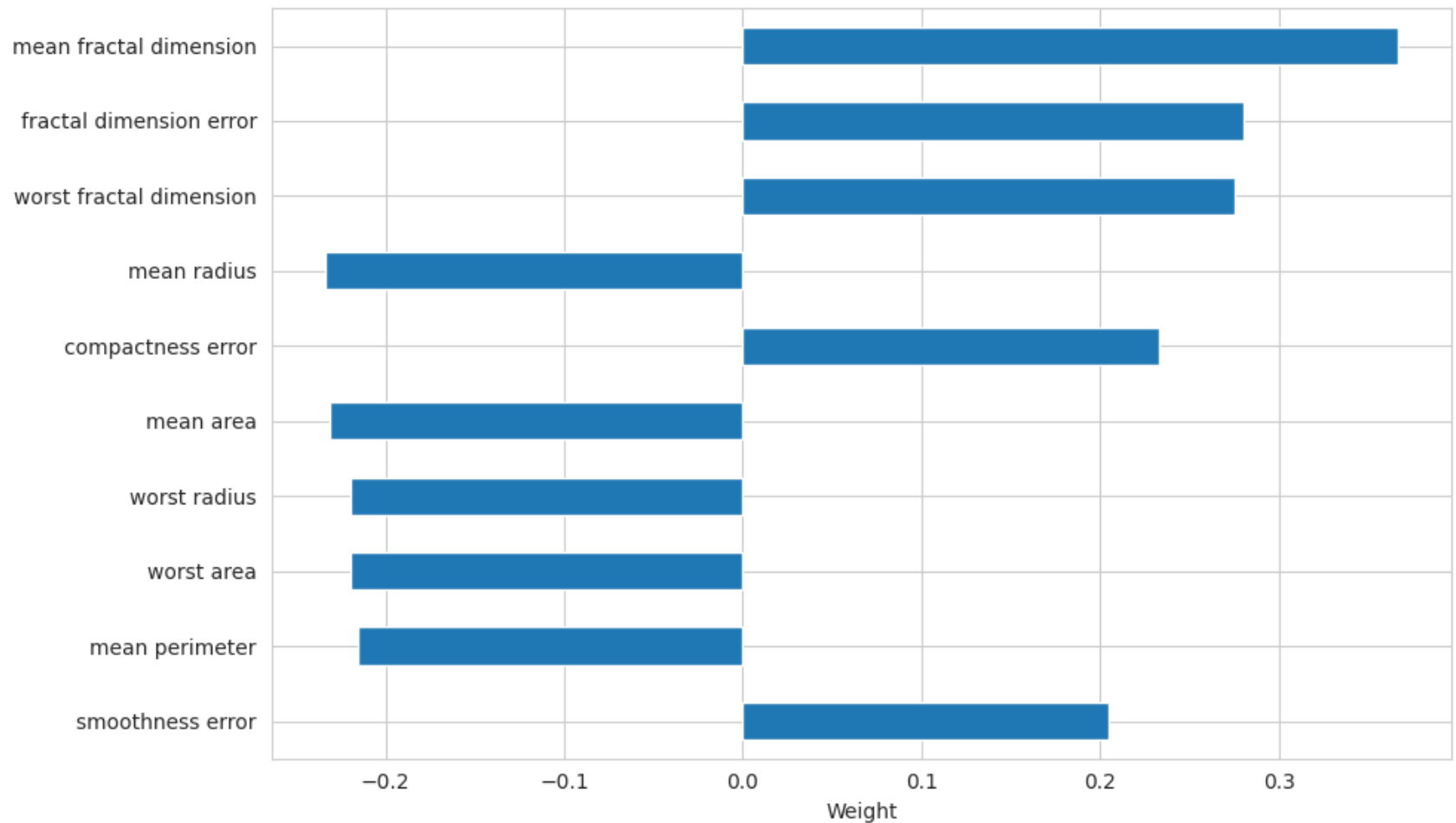
PCA:



Top Features:



Malignant vs. Benign:



ML Models:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- GridSearch (Tuning)
- Ensemble learning (Voting, Bagging, Boosting, and Stacking)

Collab Link: [here](#).

Thank you!

Questions?

